Use Cases: RAG-Diary

> Describes how each actor interacts with the system at process level, with alternative steps and failure paths, and shows the include and extend relations between use cases.

> Thai version: USE-CASES.md

1. Document control

Item	Detail
Document	Use Cases
Version	1.0
Prepared	15 October 2026
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Status	Reviewed, round 1 (25 October 2026)
Related documents	SRS, USER-STORIES, PROCESS-FLOWS, TEST-PLAN

2. Purpose and scope

This document answers one question: what does each user do with the system, and where are the alternatives and the failure points? The team uses it to design screens and write the test set.

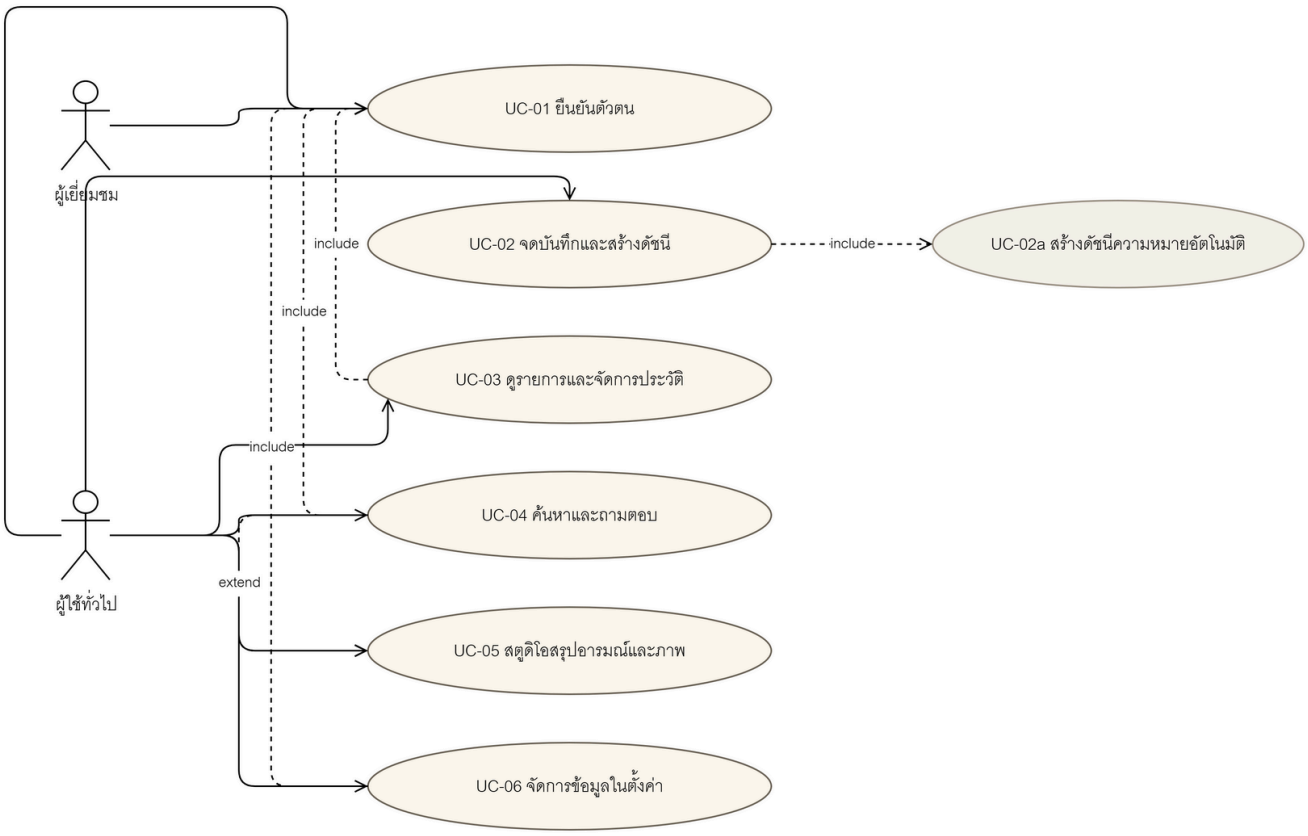
Scope: 6 use cases covering all 15 functional requirements. Out of scope: enterprise-level business process sequence (in PROCESS-FLOWS) and object-level message sequence (in SEQUENCE-DIAGRAMS).

3. Actors

Actor	Type	Description
Visitor	Primary	Someone with no account, or not signed in
Registered user	Primary	Someone with an account, working on their own entries
Authentication provider	Supporting	The service that manages accounts and sessions
Embedding provider	Supporting	The service that builds embeddings and runs search

Actor	Type	Description
Image and answer provider	Supporting	The service that generates text and illustrations

4. Use case diagram



5. Summary table

ID	Name	Primary actor	Related requirements
UC-01	Authenticate	Visitor	FR-01, FR-02, FR-03
UC-02	Write an entry and build the index	Registered user	FR-04, FR-05
UC-02a	Build the semantic index automatically (included use case)	System (called by UC-02)	FR-05
UC-03	View the list and manage history	Registered user	FR-06, FR-07, FR-08
UC-04	Search and ask questions	Registered user	FR-09, FR-10
UC-05	Mood and image studio	Registered user	FR-11, FR-12
UC-06	Manage data in settings	Registered user	FR-13, FR-14, FR-15

6. Use case detail

UC-01 Authenticate

Item	Detail
Primary actor	Visitor
Secondary actor	Registered user (for logout only)
Supporting actor	Authentication provider
Goal	Let a user reach their own entries safely
Trigger	A visitor wants to write or read a private entry
Precondition	No session yet (entered from any page)
Postcondition	The user has a working session, or sees an understandable refusal
Related requirements	FR-01, FR-02, FR-03

Main flow

1. The visitor chooses sign-up or login.
2. They enter an email and password.
3. The system validates the input.
4. The system creates a session and redirects to the home page.

Alternative flow

- Step 1: the visitor changes their mind between the sign-up and login pages; the system switches immediately.
- Step 4: the user chooses to log out; the system clears the session and redirects to the login page.

Exception flow

- Step 3: the password is wrong; the system reports invalid credentials and allows a retry.
- Step 3: the email is already registered; the system says the email is in use.
- Step 3: the network fails; the system shows an error and allows a retry.

UC-02 Write an entry and build the index

Item	Detail
Primary actor	Registered user
Supporting actor	Embedding provider
Goal	Let the user record a story and have the system prepare it for later retrieval
Trigger	The user has something to write and wants to find it again later

Item	Detail
Precondition	The user is signed in
Postcondition	The entry is stored and its semantic index is ready
Related requirements	FR-04, FR-05
Relation	Includes UC-02a, build the semantic index automatically

Main flow

1. The user chooses to write a new entry.
2. They type the content and choose or create a tag.
3. The user saves.
4. The system stores the entry in the user's account.
5. The system builds an embedding from the content automatically.
6. The system shows the new entry on the home page.

Alternative flow

- Step 2: the user is unsure of the content, cancels before saving, and comes back later.
- Step 5: the user edits the entry later; the system rebuilds the index from the new text.

Exception flow

- Step 4: the user is not signed in; the system redirects to the login page.
- Step 5: the embedding provider fails; the system notifies the user and allows a retry.

UC-03 View the list and manage history

Item	Detail
Primary actor	Registered user
Supporting actor	Authentication provider
Goal	Show the user all of their entries and let them edit or delete
Trigger	The user wants to review or correct an earlier entry
Precondition	The user is signed in
Postcondition	The list and the data reflect the latest action
Related requirements	FR-06, FR-07, FR-08
Relation	Includes authentication

Main flow

1. The user opens the list.

2. The system shows only that user's entries, newest first.
3. The user selects an entry to read it in full.
4. The user chooses to edit or delete.
5. The system stores the change.

Alternative flow

- Step 3: the user chooses edit; the system opens the editor with the current text.
- Step 3: the user chooses delete; the system asks for confirmation first.
- Step 4: the user cancels the deletion; nothing is deleted and the list stays as it was.

Exception flow

- Step 5: the deletion fails; the system shows a message and allows a retry.

UC-04 Search and ask questions

Item	Detail
Primary actor	Registered user
Supporting actor	Embedding provider, image and answer provider
Goal	Let the user retrieve entries by meaning and ask questions of their own entries
Trigger	The user cannot recall the words, or wants a summary answer from what they wrote
Precondition	The user is signed in and has at least one entry
Postcondition	The user has retrieval results or an answer grounded in their own entries
Related requirements	FR-09, FR-10
Relation	Includes authentication; extends to UC-05 when the user asks for a mood summary

Main flow

1. The user types a query or a question.
2. The system turns the text into a vector and searches that user's index.
3. The system shows results, or an answer with references to the entries.
4. The user selects a result to open the full entry.

Alternative flow

- Step 3: the user switches to question answering; the system produces an answer through the image and answer provider.

- Step 3: the user asks for a mood summary of the results; the system moves to UC-05.

Exception flow

- Step 2: no related result is found; the system says so and suggests trying again.
- Step 2: the provider fails; the system shows an understandable error and allows a retry.

UC-05 Mood and image studio

Item	Detail
Primary actor	Registered user
Supporting actor	Image and answer provider
Goal	Show the user their mood trend and create an illustration from the entries
Trigger	The user wants to review how they felt and keep an image of the memory
Precondition	The user is signed in and has at least one entry
Postcondition	The user has a summary, and possibly a generated illustration
Related requirements	FR-11, FR-12
Relation	Extends UC-04 (no direct include, because sign-in is already a precondition)

Main flow

1. The user selects entries or a date range.
2. The system produces a mood summary and an image prompt.
3. The user asks for an image from the prompt.
4. The system calls the image provider and shows the result.

Alternative flow

- Step 2: the user edits the prompt before generating; the system uses the new prompt.
- Step 3: the user generates again; the system produces a new image from the latest prompt.

Exception flow

- Step 2: the user has no entries; the system asks them to add entries first.
- Step 4: the image provider is slow or fails; the system shows status and allows a retry.

UC-06 Manage data in settings

Item	Detail
Primary actor	Registered user

Item	Detail
Supporting actor	Authentication provider
Goal	Let the user control their own data through export, deletion, and confidence that it does not leak
Trigger	The user cares about privacy and wants to manage their data directly
Precondition	The user is signed in
Postcondition	Data is exported or deleted as chosen, and isolation still holds
Related requirements	FR-13, FR-14, FR-15
Relation	Includes authentication

Main flow

1. The user opens the settings page.
2. They choose to export data or delete all data.
3. The system performs the choice.
4. The system confirms the result.

Alternative flow

- Step 2: the user chooses export; the system prepares a file with every entry.
- Step 2: the user chooses delete; the system asks for confirmation first.
- Step 3: the user cancels the deletion; nothing is deleted and all data remains.

Exception flow

- Step 3: the deletion fails; the system shows a message and allows a retry.
- Step 3: a cross-account request arrives; the system refuses and returns nothing (FR-15).

7. Include and extend relations

Type	From	To	Reason
include	UC-02	UC-02a Build the semantic index automatically	It must happen every time a new entry is written
include	UC-03	UC-01 Authenticate	Data access requires authentication first
include	UC-04	UC-01 Authenticate	Data access requires authentication first

Type	From	To	Reason
include	UC-06	UC-01 Authenticate	Data management requires authentication first
extend	UC-04	UC-05	A user may ask for a mood summary after seeing search results

> Note: UC-02 and UC-05 run after the user has signed in, which is stated as a precondition, so the diagram omits their include to UC-01 to avoid duplicate lines.

8. References

Document	Relationship
SRS	The functional requirements each UC references
USER-STORIES	The user view that corresponds to each UC
PROCESS-FLOWS	The enterprise-level process models
TEST-PLAN	Uses the flows as the basis of the test set